

ABINAYA SRIKANTH

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Mechanical Engineer | Design & Testing | Simulation & Data Analysis | Published Researcher

EDUCATION

Master of Science in Mechanical Engineering | GPA: 3.4/4.0

Expected May 2026

The University of Texas at Dallas, Richardson, TX

Coursework: Vehicle Dynamics & Control, Engineering Optimization, Reliability-Based Design, Systems Modeling & Simulation, CAD, Dynamics of Complex Networks

Bachelor of Technology in Mechanical Engineering | GPA: 3.2/4.0

May 2024

PES University, Bangalore, India

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | TÜV Rheinland India Pvt. Ltd.

Bangalore, India

January 2024 – May 2024

- Validated 45+ EV battery management systems through functional, environmental stress, thermal cycling, and durability testing, achieving 100% compliance with ISO, IEC, and UN ECE certification standards
- Configured data acquisition systems with voltage sensors, current sensors, thermocouples, and pressure transducers, capturing real-time performance data across 12+ operating conditions
- Identified 3 recurring failure modes through statistical analysis of large test datasets in MATLAB and Python (pandas, numpy), delivering root cause analysis and corrective action recommendations
- Authored 20+ technical test reports documenting methodologies, tolerancing analysis, and certification findings; interfaced with suppliers and client engineering teams on specifications and deliverables
- Improved testing throughput by 25% by redesigning workflows, automating data collection in Python, and optimizing fixture configurations

Graduate Research Engineer | The University of Texas at Dallas

Richardson, TX

August 2024 – December 2025

- Developed and validated Data-Enabled Predictive Control (DeePC) algorithms for V2X vehicular communication using MATLAB/Simulink, achieving 94% prediction accuracy (R^2) across 150+ test scenarios with 715 training samples
- Achieved 72% channel congestion reduction and 69% latency improvement through closed-loop simulation-to-test correlation and iterative optimization across 800+ simulation runs
- Built automated data processing pipelines in Python for statistical benchmarking and performance validation, reducing manual analysis time by 40%

Mechanical Design & Test Engineer | PES University

Bangalore, India

January 2023 – April 2023

- Designed braking system for tadpole three-wheeler in CATIA V5 and SolidWorks with GD&T per ASME Y14.5, reducing part count by 34% (47 to 31 components) through DFM optimization
- Built 3-DOF Simulink model for braking dynamics, validated against physical testbed: 5.4 m/s² peak deceleration, 27.1m stopping distance at 60 km/h
- Developed test procedures for brake force measurement, deceleration profiling, and stopping distance evaluation using instrumentation and data acquisition equipment

PROJECTS

Hydrogen Fuel Pressure Regulator Redesign | UT Dallas – MECH 6303 Advanced CAD

Fall 2024

- Redesigned fuel regulator housing and components in SolidWorks for hydrogen compatibility: modified body geometry, diaphragm material, seals, and pressure relief valve; applied DFM for injection molding and machining
- Planned Ansys FEA structural analysis and thermal validation (50–225°F) to qualify design under static and dynamic pressure loading

TECHNICAL SKILLS

CAD & Design: CATIA V5 (proficient), SolidWorks (proficient), AutoCAD, GD&T per ASME Y14.5, tolerance analysis, DFM/DFA, engineering drawings

Simulation & Analysis: MATLAB/Simulink (proficient), Ansys FEA, dynamic system modeling, closed-loop control, simulation-to-test correlation, predictive modeling

Testing & Manufacturing: Product validation, environmental/thermal/durability testing, DAQ systems, sensor instrumentation, root cause analysis, fixture design, process optimization, supplier interface

Programming: Python (pandas, numpy, matplotlib), MATLAB scripting, statistical analysis, automated data pipelines, Excel (advanced)

Standards: ISO, IEC, SAE, ASTM, UN ECE, ASME Y14.5, technical documentation, regulatory compliance, cross-functional collaboration

PUBLICATION

Sahani, M., Keerthi, B., **Srikanth, A.**, Mandara, B.M., Patil, S.S. (2025). Braking Dynamics of a Tadpole Type Three-Wheeler. Materials Research Proceedings, Vol. 49, pp. 366-377. DOI: 10.21741/9781644903438-37. Presented at ISME-2024, New Delhi.